

# MS&E 228: Applied Causal Inference Powered by ML and AI

## Lecture 12: Non-Parametric Structural Equations and Directed Acyclic Graphs

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**Readings:** *Applied Causal Inference Powered by ML and AI*, §7, §11.

## Where We Are

We have completed a long journey through causal machine learning estimation:

- ▶ **Potential outcomes framework:**  $Y(1)$ ,  $Y(0)$ , ATE, ATT, CATE.
- ▶ **Key identification assumption:** Conditional exogeneity  $Y(d) \perp\!\!\!\perp D \mid X$ .
- ▶ **Two identification strategies:** Conditioning & propensities.
- ▶ **Estimation paradigms:** DR/DML estimators; DR-Learner and R-Learner for CATE.
- ▶ **Overarching principle:** “Insensitive estimation formulas” robust to nuisance model errors.

*We emerged victorious from causal ML estimation! Now we switch gears completely.*

# The Big Question

We go back to the basics — the real “Causal Inference.”

## The Motivating Question

The main assumption that enabled identification was **conditional exogeneity**. But how can we *deduce* it from more **human-interpretable** assumptions?

Imagine going to a domain expert and asking:

*“Do you believe that your potential outcomes are conditionally independent of the treatment given the covariates?”*

Most domain experts won't even grasp what this means!

**Our goal:** Find a more **human-friendly** way of expressing assumptions on the data generating process, such that conditional exogeneity is a **conclusion** — not the premise.

# What We Will Achieve

In resolving this issue, we will achieve something far greater:

1. **A new language: Non-parametric structural equation models (SEMs)** — a richer, mechanistic foundation for the potential outcomes framework.
2. **A visual representation: Directed acyclic graphs (DAGs)** — a perfect fit for human-interpretable assumption elicitation.
3. **Interventions formalized:** A formal mathematical concept of an “intervention” that gives potential outcomes an **interventional interpretation**.
4. **Powerful tools:** The **Single World Intervention Graph (SWIG)** to deduce conditional exogeneity and to identify good vs. bad controls.

*This framework is not confined to conditional exogeneity — it will help us derive other identification strategies later on.*

# **Part I: From Domain Knowledge to Causal Graphs**

Building the Influence Graph from an Empirical Example

# The Hospital Messaging Example — Setup

Let's revisit the hospital messaging application from the in-class activity in the last lecture.

## Setting

Doctors monitor young patients with diabetes through a remote monitoring system and send messages to incentivize better blood glucose control.

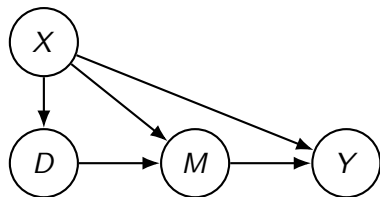
The key variables:

- ▶  $Y$  = glucose time-in-range in the days after the message
- ▶  $D$  = whether the doctor decided to send a message
- ▶  $M$  = the actual text of the message
- ▶  $X$  = all variables the doctor observes in the system (e.g., CGM data)

# The Influence Graph — Observed Variables

Here is one reasonable “influence graph” for the observed variables:

- ▶  $X \rightarrow D$ : CGM data influences the doctor's decision.
- ▶  $D \rightarrow M$ : Decision to send leads to message text.
- ▶  $X \rightarrow M$ : CGM data influences message content.
- ▶  $X \rightarrow Y$ : Patient history influences outcome.
- ▶  $M \rightarrow Y$ : The message influences glucose outcome.



*But are there other variables that influence the observed variables?*

# Adding Unobserved / Latent Variables

Yes! There are unobserved variables that influence the observed ones:

## Patient behavioral habits ( $U_p$ ):

- ▶ What they eat, exercise habits, etc.
- ▶  $U_p \rightarrow X$ : habits influence monitoring data
- ▶  $U_p \rightarrow Y$ : habits influence glucose outcome

## Doctor's capacity / workload ( $U_c$ ):

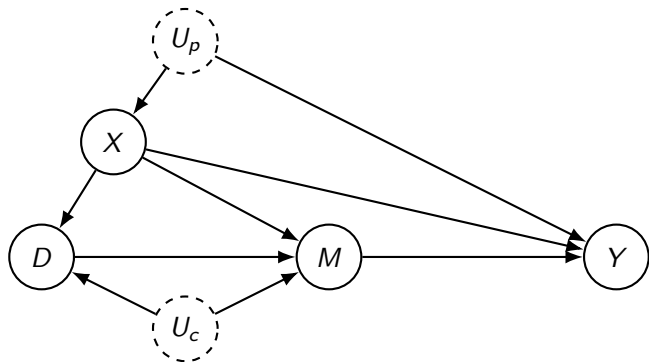
- ▶  $U_c \rightarrow D$ : workload affects decision to send
- ▶  $U_c \rightarrow M$ : workload affects message quality

## Noise terms ( $\varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$ ):

Each only influences its own node. We **omit these from the graph** for clarity and always keep in mind that each node has an associated independent noise term.



# The Complete DAG



This is the **human-interpretable assumption** we are willing to make. This graph describes how we believe, after conversing with a domain expert, the data we observe are generated.

*Dashed circles = unobserved. Each node also has implicit independent noise term  $\varepsilon_i$ .*

# The DAG as a Bayesian Network

This graph is a **Bayesian network** for the data:

## Markov Factorization

The probability law of the data factorizes according to the graph:

$$p(\mathbf{v}) = \prod_i p(v_i \mid v_{\text{Parents}(i)})$$

## What does this mean?

- ▶ Each variable is conditionally independent of its non-descendants given its parents.
- ▶ The graph captures **all conditional independencies** that hold in the data.

*In our example:*

$$p(u_p, u_c, x, d, m, y) = p(u_p) p(u_c) p(x \mid u_p) p(d \mid x, u_c) p(m \mid d, x, u_c) p(y \mid m, x, u_p)$$

# **Part II: Non-Parametric Structural Equations**

What the DAG Really Means Mathematically

# From DAG to Structural Equations

The DAG means: every variable is generated as a **function of its parent variables** and an **exogenous noise term**:

## Non-Parametric Structural Equation Model (SEM)

$$X = f_X(U_p, \varepsilon_X),$$

$$D = f_D(X, U_c, \varepsilon_D),$$

$$M = f_M(D, X, U_c, \varepsilon_M),$$

$$Y = f_Y(M, X, U_p, \varepsilon_Y)$$

- ▶ **No restriction** on how these functions look like (non-parametric!).
- ▶ The **crucial restriction** is on *what inputs* each function takes — only the **parents** of the node in the graph.
- ▶ Noise terms  $\varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$  and latent factors  $U_p, U_c$  are **jointly independent**.

# The Data Generating Process

The SEM defines a **sequential data generating process**:

1. **Draw** the jointly independent exogenous variables:  
 $U_p, U_c, \varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y \sim$  some exogenous distribution.
2. **Generate**  $X \leftarrow f_X(U_p, \varepsilon_X)$
3. **Generate**  $D \leftarrow f_D(X, U_c, \varepsilon_D)$
4. **Generate**  $M \leftarrow f_M(D, X, U_c, \varepsilon_M)$
5. **Generate**  $Y \leftarrow f_Y(M, X, U_p, \varepsilon_Y)$

## Key Insight

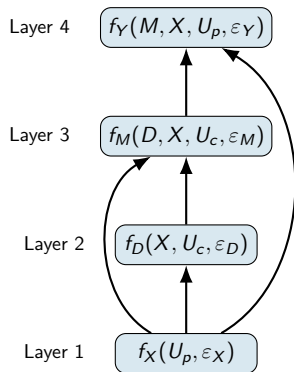
The DAG that we depicted from domain assumptions implicitly defines a **generative model** for the data!

## Side Note: Connection to Deep Generative Models

There is an interesting new line of work that takes this viewpoint and connects it to **deep generative models**:

- ▶ Each structural equation  $f_i$  can be modeled as a **neural network layer**.
- ▶ The SEM becomes a deep generative model.
- ▶ Causal inference reduces to training deep generative models with gradient descent!

See: *Xia et al. (2021). "The Causal-Neural Connection."*



## Poll 1: Understanding the SEM

### Poll Question

In the hospital messaging SEM, which of the following is the **crucial restriction** imposed by the DAG?

- (a) The structural equation functions  $f_i$  must be linear.
- (b) Each function  $f_i$  can only take the **parents** of node  $i$  as inputs.
- (c) The noise terms  $\varepsilon_i$  must be Gaussian.
- (d) The latent factors  $U_p, U_c$  must be correlated.

## **Part III: Interventions & Potential Outcomes**

What Does “Effect of Sending a Message” Really Mean?



# What Is the “Effect of Sending a Message”?

Suppose we want to measure the **effect of sending a message** (any message).

In the language of potential outcomes:

$$Y(1) = \text{outcome had we } \textit{sent} \text{ a message} \qquad Y(0) = \text{outcome had we } \textit{not}.$$

But isn't this a bit too abstract? **How do we map this to the DGP we just described?**

## A New Interpretation

Potential outcomes are the results of **interventions**: suppose we *intervened* and **forced** physicians to send a message (or not). Then:

- ▶  $Y(1)$ : the **interventional outcome** when we force  $D = 1$
- ▶  $Y(0)$ : the **interventional outcome** when we force  $D = 0$

$$\text{ATE} = \mathbb{E}[Y(1) - Y(0)]$$

# Formalizing Interventions in the SEM

What does it **formally mean** to intervene? We go to the treatment variable  $D$  and instead of its natural value, we **fix** it to some value  $d \in \{0, 1\}$ .

## DGP Under Intervention $\text{fix}(D = d)$

1. Draw  $U_p, U_c, \varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$  from **same distribution** as the factual world.
2. Generate  $X \leftarrow f_X(U_p, \varepsilon_X)$  *(unchanged)*
3. Generate  $D \leftarrow f_D(X, U_c, \varepsilon_D)$  *(natural value, still generated)*
4. But use the **fixed** value  $d$  downstream:
5. Generate  $M(d) \leftarrow f_M(\mathbf{d}, X, U_c, \varepsilon_M)$
6. Generate  $Y(d) \leftarrow f_Y(M(d), X, U_p, \varepsilon_Y)$

**Key:** We still generate  $D$  naturally (we need it in the SWIG), but every downstream equation uses the **fixed** value  $d$  instead.

# Key Insights from the Interventional DGP

1. **Formal definition of POs:** We have given a formal mathematical definition of  $Y(1)$  and  $Y(0)$  using the *exact same building blocks* as the observed world.
2. **Consistency property:** If we set  $d = D$ , we recover the factual world:

$$Y = Y(D)$$

3. **Invariance:** The only things that remain invariant across interventions are the distribution of the noise terms/latent factors and the structural equations.

## Bottom Line

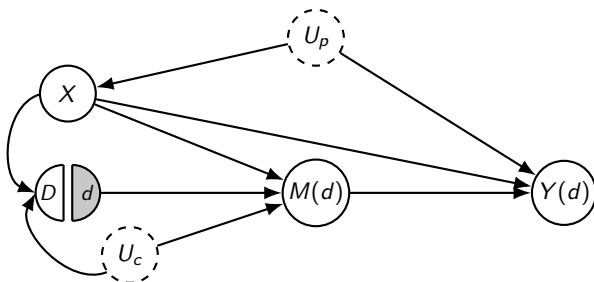
We have given an **interventional foundation** to the potential outcomes!

# The Single World Intervention Graph (SWIG)

The interventional world under  $\text{fix}(D = d)$  can be depicted by a new Bayesian network:

## SWIG Construction: "Strike Lightning on Node $D$ "

Split  $D$  into two halves: **(i)** left semi-circle  $D$  (original) receives all **incoming** edges; **(ii)** right semi-circle  $d$  (fixed, shaded) sends all **outgoing** edges. Every variable **downstream** of the fixed value becomes an interventional random variable.



# Checking Conditional Exogeneity on the SWIG

The SWIG is a **Bayesian network** for the interventional world. It contains both:

- ▶  $D$  — the natural treatment that *would have been* assigned
- ▶  $Y(d)$  — the counterfactual outcome

So we can check **conditional exogeneity** by checking conditional independence in this Bayesian network!

What we need to check

In the SWIG, does conditioning on  $X$  block all paths between  $D$  and  $Y(d)$ ?

$$Y(d) \perp\!\!\!\perp D \mid X \quad ???$$

But first — how do we check conditional independence in a Bayesian network?

# d-Separation: The Graphical Independence Criterion

## Definition: d-Separation

Two nodes  $A$  and  $B$  are **d-separated** given a set  $\mathbf{S}$  if **every** undirected path between  $A$  and  $B$  is **blocked**. A path is blocked if it contains:

1. A **chain**  $\dots \rightarrow W \rightarrow \dots$  or **fork**  $\dots \leftarrow W \rightarrow \dots$  where  $W \in \mathbf{S}$ , **or**
2. A **collider**  $\dots \rightarrow W \leftarrow \dots$  where  $W \notin \mathbf{S}$  and no descendant of  $W$  is in  $\mathbf{S}$ .

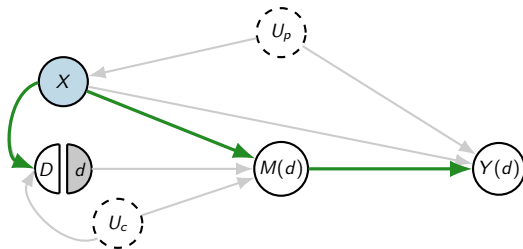
## Key Property

In a Bayesian network, if  $A$  and  $B$  are d-separated given  $\mathbf{S}$ , then  $A \perp\!\!\!\perp B \mid \mathbf{S}$ .

**Strategy:** To check  $Y(d) \perp\!\!\!\perp D \mid X$  on the SWIG, enumerate all paths between  $D$  and  $Y(d)$  and check whether each is blocked by conditioning on  $X$ .

## Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

Conditioning on  $\mathbf{S} = \{X\}$ . We highlight each path between  $D$  and  $Y(d)$ :

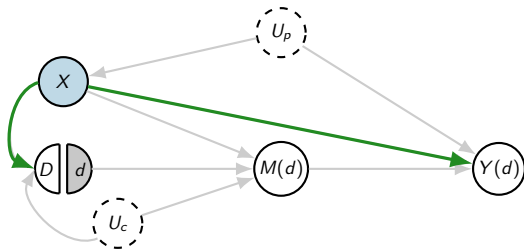


**Path 1:**  $D \leftarrow X \rightarrow M(d) \rightarrow Y(d)$

**Blocked** — fork at  $X \in \mathbf{S}$

## Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

Conditioning on  $\mathbf{S} = \{X\}$ . We highlight each path between  $D$  and  $Y(d)$ :



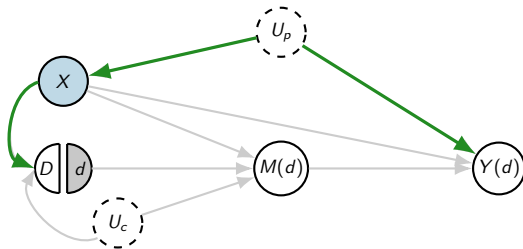
**Path 2:**  $D \leftarrow X \rightarrow Y(d)$

**Blocked** — fork at  $X \in \mathbf{S}$



## Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

Conditioning on  $\mathbf{S} = \{X\}$ . We highlight each path between  $D$  and  $Y(d)$ :

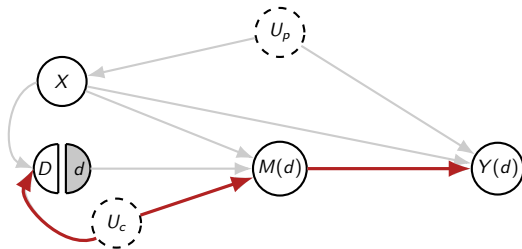


**Path 3:**  $D \leftarrow X \leftarrow U_p \rightarrow Y(d)$

**Blocked** — chain at  $X \in \mathbf{S}$

## Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

Conditioning on  $\mathbf{S} = \{X\}$ . We highlight each path between  $D$  and  $Y(d)$ :

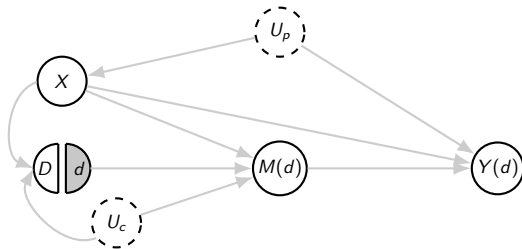


**Path 4:**  $D \leftarrow U_c \rightarrow M(d) \rightarrow Y(d)$

**Open!** —  $U_c$  unobserved,  $X$  not on path

## Checking Paths: $Y(d) \perp\!\!\!\perp D \mid X$ on the SWIG

Conditioning on  $\mathbf{S} = \{X\}$ . We highlight each path between  $D$  and  $Y(d)$ :



### Conclusion

$Y(d) \not\perp\!\!\!\perp D \mid X$ . Conditional exogeneity **fails!** The open path  $D \leftarrow U_c \rightarrow M(d) \rightarrow Y(d)$  cannot be blocked because  $U_c$  is **unobserved**.

# Why Does This Fail? — The Intuition

## The Subtle Confounding Through the Mediator

The SWIG reveals a subtle but critical issue:

- ▶  $U_c$  (doctor's workload/capacity) affects **both** the decision  $D$  **and** the message quality  $M$ .
- ▶ When we only intervene on  $D$  (force everyone to send a message), the **quality** of  $M(d)$  is still influenced by  $U_c$ .
- ▶ This creates a backdoor path:  $D \leftarrow U_c \rightarrow M(d) \rightarrow Y(d)$ .

### In plain language:

- ▶ In the observed data, doctors who *chose* to send messages typically had **more time available**, so they sent **better, more effective** messages.
- ▶ If we force *all* doctors to send messages, time-limited doctors would send **poor quality** messages.
- ▶ The observed efficacy of “sending a message” **overestimates** the true causal effect of  $\text{fix}(D = 1)$ .

## A Better Intervention: $\text{fix}(M = m)$

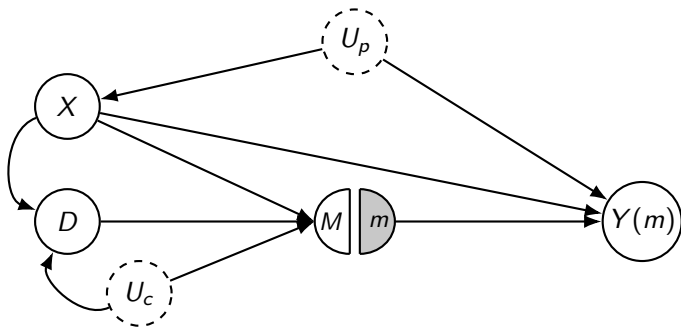
Since the problem is confounding through the mediator, what if we intervene on the **message itself**?

### DGP Under $\text{fix}(M = m)$

1. Draw  $U_p, U_c, \varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$  (same distribution).
2.  $X \leftarrow f_X(U_p, \varepsilon_X)$
3.  $D \leftarrow f_D(X, U_c, \varepsilon_D)$  (natural value)
4.  $M \leftarrow f_M(D, X, U_c, \varepsilon_M)$  (natural value, still generated)
5. But use the **fixed message**  $m$  downstream:
6.  $Y(m) \leftarrow f_Y(\textcolor{red}{m}, X, U_p, \varepsilon_Y)$

**Key:** We still generate  $D$  and  $M$  naturally (we need them in the SWIG), but the outcome equation uses the **fixed** message  $m$ .

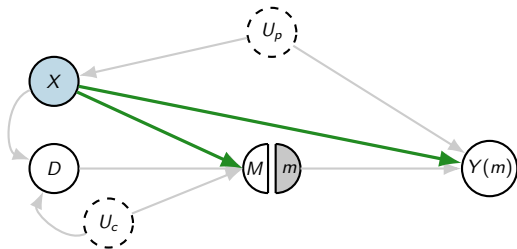
## SWIG for $\text{fix}(M = m)$



Now  $D$  is **not** split — it is a regular node. Only  $M$  is intervened upon. The outcome  $Y(m)$  depends on the fixed message  $m$ , not on the natural  $M$ .

## Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on  $\mathbf{S} = \{X\}$ . We highlight each path between  $M$  and  $Y(m)$ :

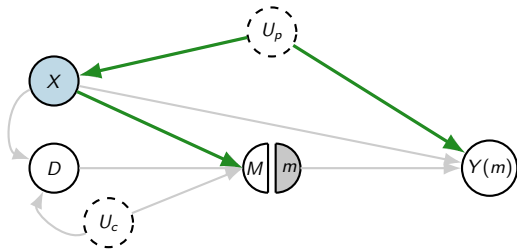


**Path 1:**  $M \leftarrow X \rightarrow Y(m)$

**Blocked** — fork at  $X \in \mathbf{S}$

## Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on  $\mathbf{S} = \{X\}$ . We highlight each path between  $M$  and  $Y(m)$ :



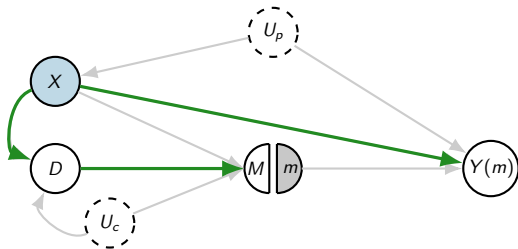
**Path 2:**  $M \leftarrow X \leftarrow U_p \rightarrow Y(m)$

**Blocked** — chain at  $X \in \mathbf{S}$



## Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on  $\mathbf{S} = \{X\}$ . We highlight each path between  $M$  and  $Y(m)$ :

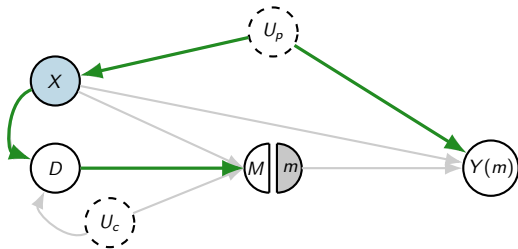


**Path 3:**  $M \leftarrow D \leftarrow X \rightarrow Y(m)$

**Blocked** — chain at  $X \in \mathbf{S}$

## Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on  $\mathbf{S} = \{X\}$ . We highlight each path between  $M$  and  $Y(m)$ :

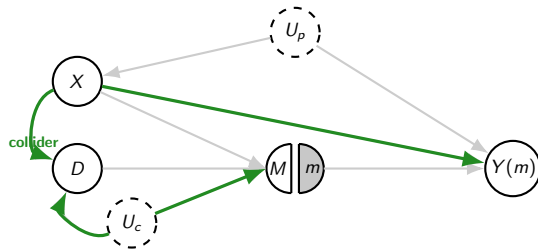


**Path 4:**  $M \leftarrow D \leftarrow X \leftarrow U_p \rightarrow Y(m)$

**Blocked** — chain at  $X \in \mathbf{S}$

## Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on  $\mathbf{S} = \{X\}$ . We highlight each path between  $M$  and  $Y(m)$ :

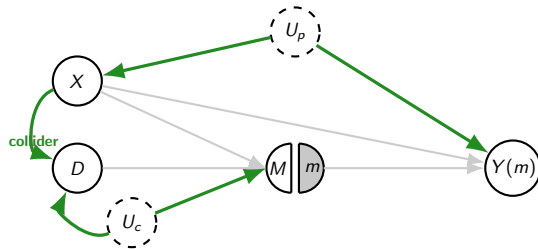


**Path 5:**  $M \leftarrow U_c \rightarrow D \leftarrow X \rightarrow Y(m)$

**Blocked** — collider at  $D$ ,  $D \notin \mathbf{S}$

## Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on  $\mathbf{S} = \{X\}$ . We highlight each path between  $M$  and  $Y(m)$ :

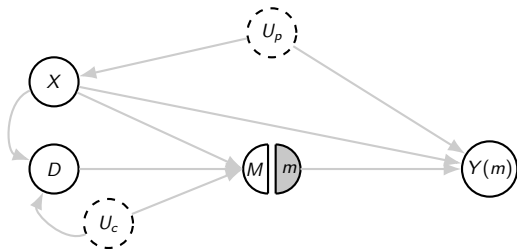


**Path 6:**  $M \leftarrow U_c \rightarrow D \leftarrow X \leftarrow U_p \rightarrow Y(m)$

**Blocked** — collider at  $D$

## Checking Paths: $Y(m) \perp\!\!\!\perp M \mid X$ on the SWIG

Conditioning on  $\mathbf{S} = \{X\}$ . We highlight each path between  $M$  and  $Y(m)$ :



### All Paths Blocked!

Every path from  $M$  to  $Y(m)$  is blocked by  $X$ . Paths through  $U_c$  hit the **collider** at  $D$  ( $U_c \rightarrow D \leftarrow X$ ), which blocks them without conditioning.

## Conditional Exogeneity **Holds** for $\text{fix}(M = m)$ !

### Result

$$Y(m) \perp\!\!\!\perp M \mid X \quad \checkmark$$

Conditional exogeneity holds when we intervene on the **message**  $M$ . We can apply all identification and estimation strategies from prior lectures to estimate the causal effect of the message content!

**Voilà!** We have deduced conditional exogeneity as a **conclusion**, where the only assumption was the **human-interpretable DAG**.

*But note: we could **not** have deduced this for the binary decision  $D$  alone — the SWIG revealed the subtle confounding through the mediator!*

# Intervening on Multiple Variables

We can also intervene on **both**  $D$  and  $M$  simultaneously:

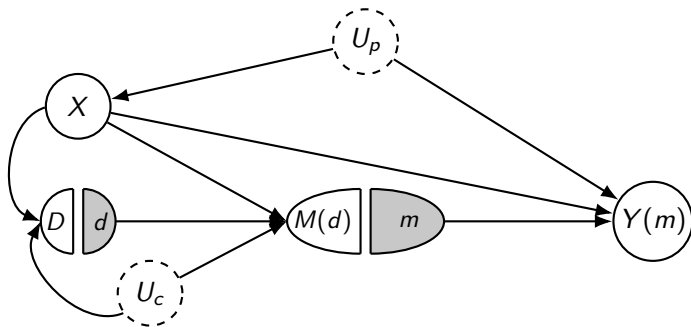
DGP Under  $\text{fix}(D = d, M = m)$

1. Draw  $U_p, U_c, \varepsilon_X, \varepsilon_D, \varepsilon_M, \varepsilon_Y$  (same distribution).
2.  $X \leftarrow f_X(U_p, \varepsilon_X)$
3.  $D \leftarrow f_D(X, U_c, \varepsilon_D)$  (natural value)
4.  $M(d) \leftarrow f_M(\textcolor{red}{d}, X, U_c, \varepsilon_M)$  (use fixed  $d$ )
5. But instead of  $M(d)$ , use the **fixed message**  $m$ :
6.  $Y(d, m) \leftarrow f_Y(\textcolor{red}{m}, X, U_p, \varepsilon_Y)$

**Notice:**  $Y(d, m) = Y(m)$  since  $d$  doesn't directly enter  $f_Y$ .

## SWIG for Double Intervention

Now we are “striking two lightnings” — intervening on both  $D$  and  $M$ :



We cannot check the condition  $Y(m) \perp\!\!\!\perp (D, M) \mid X$  on this SWIG!

**Note:** On this SWIG,  $M$  does not appear as a standalone variable — only  $M(d)$  (left half) and  $m$  (right half).

*If you cannot verify it on the SWIG you should not use it for identification!*



# What We've Achieved So Far

Through the hospital messaging example, we have:

1. **Started** from interpretable domain assumptions → **DAG**
2. **Formalized** what the DAG means mathematically → **Non-Parametric SEM**
3. **Defined** what an intervention is and the DGP under intervention
4. **Given** an interventional interpretation of potential outcomes and ATE
5. **Used** the SWIG to check conditional exogeneity — and discovered it **fails** for  $\text{fix}(D = d)$  but **holds** for  $\text{fix}(M = m)$ !

The **only assumption** was the human-interpretable DAG. Conditional exogeneity was a **conclusion** to check, not a premise — and the SWIG revealed a subtle failure we might have missed!

# The Power of the SWIG: Avoiding Subtle Mistakes

## The Mistake We Almost Made

Without the SWIG, we might have naively assumed  $Y(d) \perp\!\!\!\perp D \mid X$  and estimated the effect of “sending a message” ( $D$ ) by conditioning on patient data ( $X$ ). This would have been **wrong**.

## What the SWIG revealed:

- ▶ There is **implicit confounding through the mediator**  $M$ : the doctor's workload ( $U_c$ ) affects both the *decision* to send and the *quality* of the message.
- ▶ In the observed data, doctors who chose to send messages typically had **more time available**  $\Rightarrow$  they sent **better, more effective** messages.
- ▶ If we force *all* doctors to send messages, time-limited doctors would send **poor quality** messages — the observed efficacy **overestimates** the true effect.

**The fix:** Intervene on  $M$  directly, or additionally condition on physician workload ( $U_c$ ).

## Technical Parenthesis: Primitive Potential Outcomes

We can push interventions to the extreme — fix **all** variables, even the latent factors:

### Primitive Interventional Variables

$$X(u_p) \leftarrow f_X(u_p, \varepsilon_X)$$

$$D(x, u_c) \leftarrow f_D(x, u_c, \varepsilon_D)$$

$$M(d, x, u_c) \leftarrow f_M(d, x, u_c, \varepsilon_M)$$

$$Y(m, x, u_p) \leftarrow f_Y(m, x, u_p, \varepsilon_Y)$$

These are the **primitive potential outcomes** — the counterfactual outcome of a variable when we fix *all* of its parent variables.

Because the noise terms are independent, these primitive POs are **jointly independent**.

# Alternative Definition of Structural Equations: POs as Primitives

An equivalent way to define the whole framework:

**Step 1:** Nature generates all primitive POs from some exogenous distribution satisfying joint independence:

$$\{Y(m, x, u_p)\}, \{M(d, x, u_c)\}, \{D(x, u_c)\}, \{X(u_p)\}, U_p, U_c$$

**Step 2:** Then generate the observed data by successive “plugging in”:

$$X \leftarrow X(U_p), \quad D \leftarrow D(X, U_c), \quad M \leftarrow M(D, X, U_c), \quad Y \leftarrow Y(M, X, U_p)$$

# Mathematical Equivalence Between PO and SEMs

## The Two Sides of the Same Coin

The SEM framework and the potential outcomes framework are equivalent formulations. The SEM gives a **mechanistic** foundation of how the primitive counterfactuals are generated; the PO framework gives a **counterfactual** foundation.

## Two Meeting Points

Typically, we care about average **interventional counterfactuals**, not averages of these primitive potential outcomes. The interventional counterfactuals are the most natural analogue of what people typically define as the potential outcomes in the potential outcomes framework.

# Takeaways

1. **DAGs as Domain Assumptions:** A directed acyclic graph encodes human-interpretable assumptions about the data generating process. The associated **non-parametric SEM** formalizes it mathematically.
2. **Interventional Potential Outcomes:** Potential outcomes arise naturally as **interventional random variables** from the SEM. An intervention means “fixing” a variable to a value in the structural equations.
3. **SWIGs Reveal Subtle Confounding:** The SWIG showed that  $Y(d) \not\perp\!\!\!\perp D \mid X$  due to implicit confounding through the mediator, but  $Y(m) \perp\!\!\!\perp M \mid X$  holds. Without the SWIG, this subtle failure is easy to miss.
4. **Choose the Right Intervention:** The causal question matters: intervening on the **message content**  $M$  is identifiable; intervening on the **binary decision**  $D$  alone is not (without controlling for workload).

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